

PEER REVIEW FORM

SECTION 1 — PAPER IDENTIFICATION

Paper Title Three Notions of Linearization for a Discrete Ising System: Operator, Statistical, and Projection-based Approximations	Submission ID / Authors 876e
Review Date Apr, 2026	Course / Assignment Peer Review 1 st Round

SECTION 2 — PAPER SUMMARY

Demonstrate understanding of the submission in its entirety (required for Rigor score)

2.1 Problem & Motivation

Briefly describe the problem the paper addresses and why it matters. Cite specific passages if helpful.

No single form of “linearization” exists for stochastic, high-dimensional nonlinear systems. Each choice in how to linearize produces fundamentally different models with distinct properties. The paper leaves out concrete reasons we should care, presumably because readers of the paper would be familiar with the broad applicability and interest in linearization.

The problem is well described in the Background section, where notions of simplification are Discussed. There, it becomes clear why intelligent simplifying assumptions (like linearization techniques) become critical to the capacity of scientific research and the conclusions we can draw from analysis.

2.2 Core Approach / Methodology

Summarize the technical approach. Include key equations, algorithms, or frameworks referenced in the paper.

The author studies a discrete-time Ising model with Glauber dynamics to ask how three distinct linearization techniques differ in resulting stability properties, fixed points, and structure.

2.3 Hypotheses Stated by the Authors

List the specific hypotheses the paper claims to support or refute. Quote or paraphrase directly.

The paper states, “In particular, we examine how operator-based, probability-based, and coarse-grained representations differ in their ability to reproduce transient dynamics, long-time behavior, and parameter-dependent trends.” Later in the introduction it is stated, “... the remaining models are judged by how well they reproduce [the reference model’s] manifold of trajectories and summary statistics”.

No formal hypothesis is ever provided, but the author claims that an emphasis is kept on model validity domain, rather than on formal derivations of any single reduced model.

The author later states that the purpose of this paper is to “expose the tradeoffs between fidelity, interpretability, and computational cost.” While not a hypothesis in the traditional sense, it provides adequate direction to the paper.

2.4 Key Results

What are the main results? Are they quantitative? Do they directly address the hypotheses?

Stability is mentioned in the abstract but not discussed in the results or supported by evidence. This is problematic.

The title and abstract suggest 3 distinct linear representations, but then section 5 (model simplifications) and the chart of results detail 4 models apart from the reference model. All converge to similar values, differing mainly in variance and steady-state bias. It would benefit the paper greatly to see a mapping between the 5 models chosen, and which of the 3 representations they match. Is model 5 coarse-grained, probability-based, or operator-based? This is not immediately clear to the user. Though it can be deduced from the math for models 2-4, model 5 is ambiguous as it seems to be operator- and probability-based and it is unclear if model 5 is chosen because the author wanted a blend between methods or if this should be classified as one or another for the purposes of comparing classes of methods. The paper title seems to suggest that three notions will be compared, not combined and as a reader I could use some prep before having to figure out how I should treat model 5.

The results are quantitative, and nothing new jumps out at me: it is clear even before the experiment took place that different low-order approximations would not be able to retain all data from the experiment completely, though they would likely end up close to one another through the experiment if they are faithful surrogates. Shortcomings are mentioned in the results section, but it is unclear how much “compute” would be necessary to draw stronger conclusions, or how much “compute” was available to the user.

The author states that they have discussed error structure, but the single presentation of parameter magnetization trajectory with confidence intervals (especially given that not all parameter regions are shown) does not strongly match the conclusion statement claiming error structure discussions. A stronger expansion of the phrase “the steady state was also tricky to make consistent between models” would strengthen the paper significantly. Clearly work has been done, but it doesn’t seem to be presented in a way that helps me see “tractability” or “stability”.

SECTION 3 — SCORES

Circle or mark one score per row (0 = poor → 5 = excellent). Justification is required for every row.

Scoring guide:

0	1	2	3	4	5
Absent / completely inadequate	Major deficiencies	Significant issues	Adequate with weaknesses	Good, minor issues	Excellent

3.1 Introduction & Related Work

X	Does the paper clearly describe the problem setting?
X	Does the paper clearly motivate the problem?
X	Is a sufficient amount of relevant prior work included?
-	Are the paper's contributions clearly stated?
-	Is prior work sufficiently compared and contrasted with the paper's contributions?

Category	0	1	2	3	4	5	Score: 3.5
Introduction & Related Work	☐	☐	☐	X	X	☐	Prior work is described in that the derivations of the linear models for the system are derived. As mentioned previously, motivation is sufficiently fleshed out. Related work is not mentioned, nor is there a description of which models are commonly preferred for which types of work on this system. As such, it is hard to “compare” prior literature. Contributions are not clearly stated and are left vague. These could be greatly improved, and the author is encouraged to do so. Clearly the author has done really good work with this linearization, but this section can be strengthened to reflect it.

3.2 Problem Statement

X	Does the paper have a clear problem statement?
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X	Is the problem statement sufficiently mathematically formulated?
X	Is enough mathematical background given to understand the problem?
-	Does the paper clearly state one or more hypotheses?
X	Does the paper outline experiments to support or refute the hypotheses?

Category	0	1	2	3	4	5	Score: 4.5
Problem Statement	☐	☐	☐	☐	X	☐	While no direct hypothesis is included, the paper clearly states what it is trying to compare, provides great mathematical background to support it, and outlines a figure guide and interpretation to understand the major results of the paper.

3.3 Results

X	Does the paper have convincing results?
-	Are the results novel?
-	Does the paper provide convincing evidence to support or refute the hypotheses?
X	Are the simulations / experiments thorough and rigorous?

Category	0	1	2	3	4	5	Score: 3.5
Results	☐	☐	☐	X	X	☐	It is unclear if the results are novel. These linearization techniques are not compared to prior work, but because they are not mentioned as novel, it is assumed they are not. Thus, we cannot conclude that results produced are novel, but rather a summarizing/collecting of various well-known methods for comparison. This is believed to be the novelty.

3.4 Conclusions

-	Does the paper discuss how the results support or refute the hypotheses?
-	Does the paper thoroughly discuss avenues for future work?

Category	0	1	2	3	4	5	Score: 3
Conclusions	☐	☐	☐	X	☐	☐	The first time a diagonal transform is mentioned is at the end of the paper... I would have expected to hear it described earlier. Please correct me if I'm wrong but I do not see the conclusion as very strong. It states for the first time your methods and does not provide justification for the future work. Why use a genetic algorithm as opposed to anything else?

3.5 Professionalism

X	Does the paper use a standard template from a professional society?
X	Is the paper organized with standard sections?
-	Is the paper clearly written and well organized?
-	Are the contributions clearly outlined?
-	Is the hypothesis clearly stated?
X	Is the paper well-written and free of grammatical / spelling errors?
/	Are any parts of the paper low-quality AI-generated content ('AI slop')?
☐	Does the paper include a detailed disclosure on the use of AI?
-	Is the paper at least 6 pages with no more than 2 pages of references?

Category	0	1	2	3	4	5	Score: 2
Professionalism	☐	☐	X	☐	☐	☐	The appendix feels like an AI generated discourse in part, more than it feels like an appendix. With the conclusions not matching the paper, and the title and abstract claiming bins and then the models not being explicitly labeled in bins, this paper feels AI generated. I highly suggest cleaning up the wording if it is not so as to help with the clarity. No AI disclosure is included so it is assumed no AI was used, but this feels unlikely.

SCORE SUMMARY	Score (0–5)	Weight
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Introduction & Related Work	3.5 / 5	<i>Equal</i>
Problem Statement	4.5 / 5	<i>Equal</i>
Results	3.5 / 5	<i>Equal</i>
Conclusions	3 / 5	<i>Equal</i>
Professionalism	2 / 5	<i>Equal</i>
OVERALL SCORE	3.3 / 5	

SECTION 4 — DETAILED WRITTEN FEEDBACK

Your written comments are the primary basis for your Clarity and Rigor scores as a reviewer

4.1 Strengths

This paper is strong in that it has lots of potential. The mathematics for the 5 models are well explained.

4.2 AI Content Assessment

N/A

SECTION 5 — FINAL RECOMMENDATION

- ☐ Accept as-is — the paper is publication-ready.
- ☐ Accept with minor revisions — small changes needed, no re-review required.
- ☒ Major revision required — significant issues; must be re-reviewed.
- ☐ Reject — fundamental problems that cannot be resolved in revision.

5.1 Recommendation Rationale

Briefly explain your recommendation in 2–4 sentences, connecting it to the major concerns and strengths identified above.

See above.

